

Memo
EBN111 Klastoets/Class Test#1: 1 Maart/March 2013

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Dissipline / Discipline:		Taal / Language:			Afr	Eng			

Q1 [2]	Q2 [2]	Q3 [2]
$w = 297 \text{ kJ}$ ✓✓	$P_{8v} = -16 \text{ W}$ ✓ $P_{3A} = 30 \text{ W}$ ✓	$v_z = 4.8 \text{ V}$ ✓✓
Q4 [3]	Q5 [4]	Q6 [2]
$v_x = 7.5 \text{ V}$ ✓ $v_y = 6.7 \text{ V}$ ✓ $i_x = 0.83 \text{ A}$ ✓	$i_l = 0.4 \text{ A}$ ✓✓ $v_2 = -8 \text{ V}$ ✓✓	$R_{eq} = 20 \Omega$ ✓✓

Finale Punt/Final Mark:

/15

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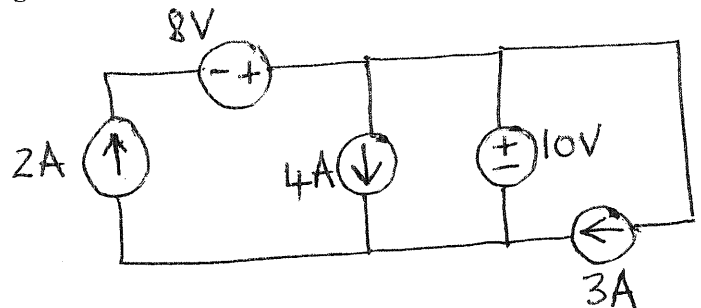
- Q1 'n Nuwe soort battery lewer 10W drywing vir 8 ure sonder enige variasie in spanning of stroom. Na 8 ure val die drywingsuitset lineêr van 10W na 0 binne 30 minute. Bepaal die energie stookapasiteit van die battery. [2]
A new type of battery delivers 10W of power for 8 hours without voltage or current fluctuation. After 8 hours the power output drops linearly from 10W to 0 in 30 minutes. Determine the energy storage capacity of the battery.

$$\begin{aligned} W &= 10 \times 8 \times 60 \times 60 + \frac{1}{2} \times 10 \times 30 \times 60 \\ &= 288000 + 9000 \\ &= 297 \text{ kJ} \checkmark \checkmark \\ &\quad \longrightarrow \end{aligned}$$

- Q2 Bereken die drywing ge-assosieer met die 8V spanningsbron en die 3A stroombron. [2]
Calculate the power associated with the 8V voltage source and the 3A current source.

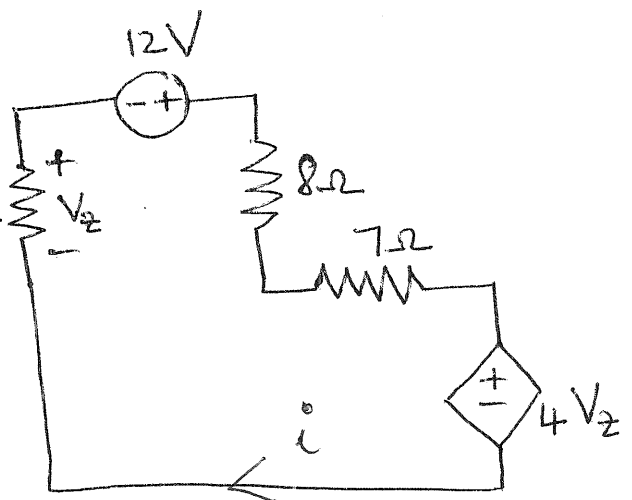
$$P_{8V} = -2 \times 8 = -16 \text{ W} \checkmark$$

$$P_{3A} = 3 \times 10 = 30 \text{ W} \checkmark$$



- Q3 Bepaal v_z in die stroombaan. [2]
Determine v_z in the circuit.

$$\begin{aligned} -V_z - 12 + 15i + 4V_z &= 0 \\ \text{waar } V_z &= -30i \\ +30i - 12 + 15i + 4(-30i) &= 0 \\ -75i &= 12 \Rightarrow i = -0.16 \text{ A} \\ V_z &= 4.8 \text{ V} \checkmark \checkmark \\ &\quad \longrightarrow \end{aligned}$$



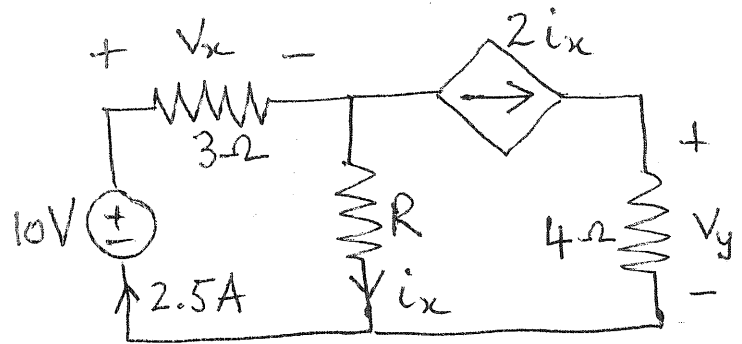
- Q4 Bereken v_x , v_y en i_x in die stroombaan. [3]
Calculate v_x , v_y and i_x in the circuit.

$$V_x = 2.5 \times 3 = 7.5 \text{ V} \checkmark$$

$$2.5 = 3i_x$$

$$\Rightarrow i_x = 0.833 \text{ A} \checkmark$$

$$V_y = 2i_x \times 4 = 6.664 \text{ V} \checkmark$$



- Q5 Bereken/Calculate

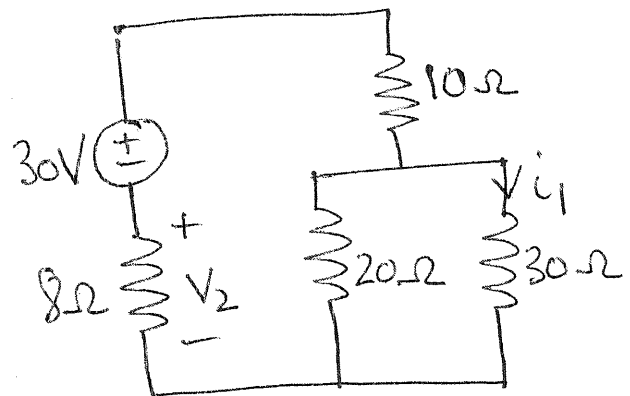
(a) i_1 [2]

(b) v_2 [2]

$$20 \parallel 30 = 12 \Omega$$

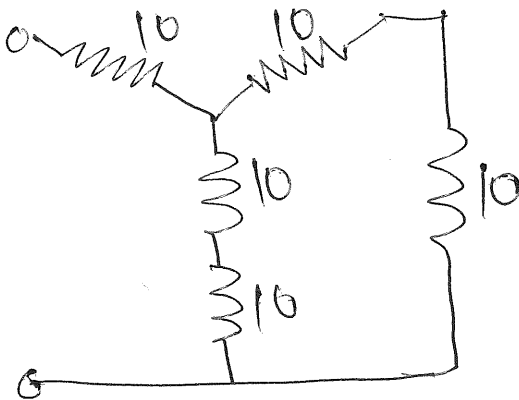
$$V_2 = - \frac{8}{8+12+10} \cdot 30$$

$$= -8 \text{ V} \checkmark \checkmark$$



$$i_1 = \frac{20}{50} \times \frac{30}{30} = 0.4 \text{ A} \checkmark \checkmark$$

- Q6 Bereken R_{eq} . [2]
Calculate R_{eq} .



$$R_{eq} = 20 \parallel 20 + 10$$

$$= 20 \Omega \checkmark \checkmark$$

